

RACHEL BOSCO

rachelbosco830@gmail.com | +91 9513689671 | github.com/rachelbosco04 | linkedin.com/in/rachelbosco | rachelbosco04.github.io

PROFESSIONAL SUMMARY

B.Tech graduate in AI & ML (CGPA: 8.7/10) with 6 months of experience building and deploying end-to-end ML systems — including a churn prediction model achieving 0.84 AUC-ROC and a PCB defect detection system reaching 96.68% accuracy. Skilled in PyTorch, LightGBM, and Scikit-learn, with hands-on experience in transfer learning, class imbalance handling (SMOTE), model explainability (SHAP, Grad-CAM), and real-time deployment via Flask REST APIs. GreenTech Hack 2024 winner.

EDUCATION

B.Tech in Computer Science (AI & ML) — CGPA: 8.7/10

2022 – 2026

Garden City University, Bengaluru

SKILLS

- **Languages:** Python, C++, SQL, HTML5, CSS3
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO, MediaPipe, Tesseract OCR
- **MLOps & Deployment:** Flask, Streamlit, REST APIs, Git/GitHub, Docker (basic), CI/CD concepts
- **Data & Databases:** PostgreSQL, Pandas, NumPy, Apache Spark (PySpark)
- **Specialisations:** Machine Learning, Computer Vision, Deep Learning, Natural Language Processing

EXPERIENCE

Data Science & ML Intern – Zaalima Development Pvt. Ltd. | Bengaluru

Jan 2026 – Apr 2026

- Developed end-to-end ML systems for predictive maintenance and telecom churn prediction using IoT sensor and customer data with LightGBM, PyTorch, and Apache Spark.
- Built and compared multiple ML models, including LightGBM, PyTorch, Logistic Regression, and Random Forest, achieving 0.84 AUC-ROC for churn prediction.
- Deployed Flask REST APIs with SHAP-based explainability and an interactive Spark dashboard for real-time risk monitoring and churn analytics.

AIML Intern – Internz Learn Pvt Ltd | Bengaluru

Oct 2025 – Jan 2026

- Developed AI-powered analytics systems including an NLP-based resume screening platform and a multi-model sales forecasting dashboard, using TF-IDF vectorization, cosine similarity, and time-series forecasting algorithms (Holt-Winters, SARIMA, Random Forest)
- Performed data preprocessing, feature engineering, and model evaluation (RMSE, MAE, MAPE); built interactive Streamlit web applications with data visualization to rank candidates and forecast retail demand with under 3% forecast error

PROJECTS

VisionSpec QC – PCB Defect Detection & Quality Inspection

- Developed an end-to-end PCB defect detection system using PyTorch and MobileNetV2 Transfer Learning, achieving 96.68% test accuracy and ~28 FPS inference; integrated Grad-CAM explainability and a Flask-based interactive dashboard for real-time industrial quality inspection.

FabricScan AI – Fabric Identifier System

- Trained an Xception-based classifier across 8 fabric categories achieving 95.8% peak accuracy using Multi-Task Learning; deployed a Streamlit app with voice assistant, 3D texture visualisation, and PostgreSQL-backed session analytics.

Blog to Podcast Agent

- Built a blog-to-podcast system using Groq LLM (Llama 3.1) and Text-to-Speech synthesis that automates article extraction, conversational summarization, and multilingual audio generation; processed 6,000+ character articles using chunk-based summarization with downloadable podcast output.

AskMyNotes – RAG-Based PDF Q&A System

- Developed an end-to-end RAG-based PDF question answering system using Python, PyPDF, Sentence-Transformers, Streamlit, and the Claude API, implementing chunking, embeddings, and cosine-similarity retrieval to generate relevant, context-grounded answers from study notes with reduced hallucinations.

ACHIEVEMENTS & CERTIFICATIONS

- **Winner – GreenTech Hack 2024** (Dr. A.P.J Abdul Kalam School of Engineering, Garden City University)
- Python Essentials – Cisco
- Mastering Big Data Analytics with PySpark – Infosys Springboard
- Hadoop Architecture & HDFS – Infosys Springboard
- Claude Code in Action – Anthropic
- Claude with Google Cloud's Vertex AI – Anthropic